Finance Architecture Audit

Version: v1.0

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Scope: Complete finance data-flow audit across UI, hooks, database, triggers and derived reporting tables.

Status: Analysis only — no business logic, schema, or calculation has been changed.

1. Executive Summary

The finance subsystem is a **client-direct PostgREST architecture**: every UI page queries Postgres tables directly via the Supabase JS client. There is **no service layer** between UI and database. All cross-table consistency is enforced by **triggers** (purchase→expense, purchase→inventory, billing payment dedupe, referral revenue, FY actuals).

What is solid

- Payment capture (`billing_payments`) is well-defined and dedupe-protected by a trigger-generated key.

- Purchases automatically mirror into `expenses` and `inventory_items` via `sync_purchase_to_expense` and `handle_purchase_inventory_update`.

- Collected = `SUM(billing_payments.amount)` by `payment_date` is the cash-basis source of truth and is consistently used in `Finance.tsx`.

- FY actuals are derived deterministically from `billings` + `expenses` via `recompute_fy_actuals`.

Top 5 risks

1. **No** `branch_id` **on any money table** (`billings`, `billing_items`, `billing_payments`, `expenses`, `purchases`, `purchase_items`, `inventory_items`, `inventory_transactions`, `deposits`, `orthodontic_payments`, `emi_insurance_cases`, `lab_orders`, `diagnostic_orders`, `referrals`). Branch filters in the UI are effectively no-ops for finance; every branch sees consolidated numbers.

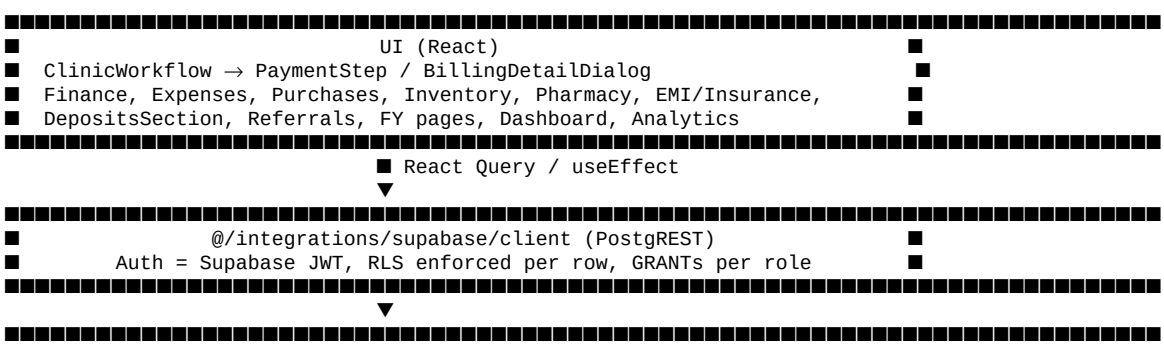
2. **No void/cancelled/refund status on** `billing_payments` — refunds must be modeled as negative `amount` rows; there is no audit trail of cancellations.

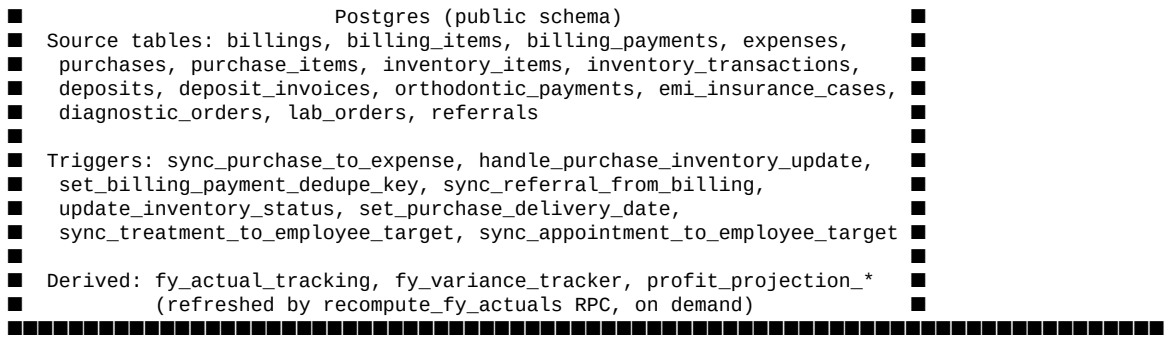
3. `recompute_fy_actuals` **is on-demand only** (called from the FY page's "Refresh" button). FY dashboards drift from live `billings`/`expenses` until a manual refresh.

4. **Dual collected-calculation paths**: `Finance.tsx` uses `billing_payments`, but `FinanceReport.tsx` falls back to `billings.amount_paid` when `billingPayments` is empty — two slightly different definitions of "Collected" coexist.

5. **Pharmacy sales are not booked into** `billings` as a distinct revenue stream; pharmacy items only flow through `inventory_items` consumption. Pharmacy revenue is therefore invisible to Finance unless billed manually.

2. Finance Architecture Overview





Key architectural notes

- **No backend API**. UI talks to PostgREST directly. Business rules live in triggers or in component code.

- **Branch scoping** exists for clinical/HR data via `BranchContext` and `user_branches`, but **money tables have no** `branch_id` **column** — branch filters on Finance are effectively cosmetic.

- **Date anchoring**: `billings.created_at`, `billing_payments.payment_date`, `expenses.expense_date`, `purchases.purchase_date`. These are the four authoritative date axes.

- **RLS**: every public table has policies. Money tables grant `SELECT/INSERT/UPDATE/DELETE` to `authenticated`.

3. Financial Data Flow

3.1 Money-entry points (where money first enters the system)

Entry point	UI surface	Writes to	Trigger side-effects
Patient pays at checkout	`PaymentStep.tsx` → `billing_payments.insert`	`billing_payments` (+ `billings.amount_paid`, `billings.payment_status` updated in same flow)	`set_billing_payment_dedupe_key`, `sync_referral_from_billing`
Invoice raised	`PaymentStep.tsx` → `billings.insert` + `billing_items.insert`	`billings`, `billing_items`	`sync_referral_from_billing`
Inventory consumed during treatment	`PaymentStep.tsx` (lines 170–204)	`inventory_items.update`, `inventory_transactions.insert` (`type='stock_out'`)	`update_inventory_status`
Purchase invoice from vendor	`Purchases.tsx` → `purchases.insert` + `purchase_items.insert`	`purchases`, `purchase_items`	`sync_purchase_to_expense` (when status=paid), `handle_purchase_inventory_update`, `set_purchase_delivery_date`
Manual expense	`Expenses.tsx` → `expenses.insert`	`expenses`	`generate_expense_number`
Bank deposit	`DepositsSection.tsx`	`deposits`, `deposit_invoices` (link rows)	—
Ortho EMI / installment	`OrthoTreatmentStep.tsx`	`orthodontic_payments`	—
EMI / Insurance disbursement	`EMIInsurance.tsx`	`emi_insurance_cases`	`generate_emi_case_code`
External diagnostic / lab cost	`DiagnosticsStep.tsx`, `LabOrders.tsx`	`diagnostic_orders.cost`, `lab_orders.rate`	—
Referral revenue attribution	(derived)	`referrals.revenue_generated`, `referrals.status`	`sync_referral_from_billing` fires from `billings` insert/update

3.2 End-to-end trace per consumer

Finance Dashboard (`src/pages/Finance.tsx`)

- Reads: `billings`, `billing_items`, `billing_payments`, `expenses`, `purchases`, `purchase_items`.

- Computes: Total Billed (sum of `billings.total_amount` in period by `created_at`), Collected (sum of `billing_payments.amount` in period by `payment_date`), Outstanding (`total_amount − amount_paid` for non-cancelled billings), Expenses, Net Profit.

Revenue — same source as Finance Dashboard; identical formulas, just rendered as charts.

Billing — `Finance.tsx` invoices table; `BillingDetailDialog.tsx` for line items.

Payments — `billing_payments` is the only authoritative cash table. Re-displayed on `Finance.tsx` (Payment Collection tab), `PatientPortal.tsx`, `HistoryStep.tsx`.

Receivables — derived (`Σ total_amount − Σ amount_paid` per billing for `payment_status IN ('pending','partial')`).

Expenses — `expenses` table. Two writers: manual entry (`Expenses.tsx`) and auto-mirror from `purchases` via `sync_purchase_to_expense`.

Vendor Payments — derived from `purchases.payment_status='paid'`. No dedicated vendor-ledger table.

Purchase Management — `purchases` + `purchase_items`. Trigger `handle_purchase_inventory_update` increments `inventory_items.quantity` and inserts an `inventory_transactions` row of `type='stock_in'`.

Inventory — `inventory_items` (current state) + `inventory_transactions` (audit log). Value = `Σ (quantity × cost_price)`.

Pharmacy — `Pharmacy.tsx` reads `inventory_items` filtered by category. **Pharmacy sales do not write to** `billings` — gap documented in §11.

Reports — `FinanceDownloadReports.tsx`, `DownloadReports.tsx`. Re-read source tables and export CSV/PDF.

Profit & Loss — computed in `Finance.tsx` and in `fy_actual_tracking.actual_profit = actual_revenue − actual_expense`.

Cash Flow — derived ad-hoc from `billing_payments` (in) + `expenses` (out). No dedicated cashflow table.

Patient History — `HistoryStep.tsx`, `PatientPortal.tsx` read `billings` + `billing_payments` for the patient.

4. Module Dependency Diagram

```
graph LR
  Workflow[Clinic Workflow] -->|writes| Billings
  Workflow -->|writes| BillingItems[Billing Items]
  Workflow -->|writes| Payments[Billing Payments]
  Workflow -->|consumes| Inventory
  Purchases -->|trigger| Expenses
  Purchases -->|trigger| Inventory
  Purchases -->|writes| PurchaseItems[Purchase Items]
  Pharmacy -->|reads| Inventory
  EMI[EMI / Insurance] -->|writes| EMICases[EMI Cases]
  Ortho[Ortho Treatment] -->|writes| OrthoPay[Ortho Payments]
  Deposits -->|writes| DepositsT[Deposits]
  Deposits -->|links| Billings
  Billings -->|trigger| Referrals
  Billings -->|RPC| FYActuals[FY Actual Tracking]
  Expenses -->|RPC| FYActuals
  FYActuals --> FYDash[FY Dashboards]
  Billings --> Finance[Finance Dashboard]
  Payments --> Finance
  Expenses --> Finance
  Purchases --> Finance
  Billings --> PatientHistory[Patient History]
  Payments --> PatientHistory
  Billings --> Analytics
  Payments --> Analytics
  Expenses --> Analytics
```

5. Database Relationship Diagram

```
erDiagram
  patients ||--o{ billings : has
  billings ||--o{ billing_items : contains
  billings ||--o{ billing_payments : "paid via"
  billings ||--o{ deposit_invoices : "deposited in"
  deposits ||--o{ deposit_invoices : groups
  patients ||--o{ orthodontic_payments : "via orthodontic_cases"
  patients ||--o{ emi_insurance_cases : has
  patients ||--o{ referrals : "referred_patient_id"
  purchases ||--o{ purchase_items : contains
  purchase_items }o--|| inventory_items : "links to"
  inventory_items ||--o{ inventory_transactions : "audited by"
  purchases ||--o| expenses : "mirrored to (trigger)"
  billings ||--o{ referrals : "drives revenue_generated (trigger)"
  treatment_plans ||--o{ treatment_plan_items : contains
  billings }o--|| treatment_plans : "plan_id"
  billings }o--|| patient_checkins : "checkin_id"
```

Read/Write matrix (money tables)

Table	Writers	Readers
`billings`	`PaymentStep`, `BillingDetailDialog`	`Finance`, `FinanceReport`, `Dashboard`, `PatientPortal`, `HistoryStep`, `Analytics`, `Referrals`, `recompute_fy_actuals`
`billing_items`	`PaymentStep`, `BillingDetailDialog`	`Finance`, `FinanceReport`, `Analytics`
`billing_payments`	`PaymentStep`	`Finance`, `FinanceReport`, `PatientPortal`, `HistoryStep`
`expenses`	`Expenses`, trigger `sync_purchase_to_expense`, trigger `generate_expense_number`	`Finance`, `FinanceReport`, `recompute_fy_actuals`
`purchases`	`Purchases`	`Finance`, `Purchases`, trigger fan-out
`purchase_items`	`Purchases`	`Finance`, `Purchases`, trigger `handle_purchase_inventory_update`
`inventory_items`	`Inventory`, `InventoryItemForm`, `PaymentStep` (consume), trigger `handle_purchase_inventory_update`	`Inventory`, `Pharmacy`, `PaymentStep`
`inventory_transactions`	`PaymentStep`, trigger	`Inventory`
`deposits` / `deposit_invoices`	`DepositsSection`	`DepositsSection`, `Finance` (Deposits tab)
`orthodontic_payments`	`OrthoTreatmentStep`	`OrthoTreatment`
`emi_insurance_cases`	`EMIInsurance`	`EMIInsurance`
`referrals`	`Referrals`, trigger `sync_referral_from_billing`	`Referrals`, `Dashboard`

6. API / Service Flow

```
User action (UI component)
        ■
        ▼
React state / form
        ■
        ▼
supabase.from("<table>").insert/update/select   ← from src/integrations/supabase/client.ts
        ■
        ▼
PostgREST (Supabase) — applies JWT + RLS policies
        ■
        ▼
Postgres table mutation
        ■
        ■■■ BEFORE/AFTER triggers (sync_purchase_to_expense, dedupe_key, etc.)
        ■■■ Cascading updates to derived tables
        ■■■ No write-ahead audit log (gap)
        ■
```

```
        ▼
UI re-fetches (React Query invalidation OR manual reload via useEffect)
        ■
        ▼
Finance Dashboard / Reports / Patient History redraw
```

Edge functions in `supabase/functions/` (`create-staff-user`, `manage-admin-users`, `meta-ads`, `platform-insights`, `smile-design-ai`, `social-inbox`, `social-reply`, `verify-platform-connection`, `whatsapp-live-insights`) — **none mutate finance tables**. Finance has no edge-function surface today.

RPC: `recompute_fy_actuals(p_branch_id, p_financial_year, p_month)` called from `useFYData.refreshActuals`.

7. Rendering Matrix

Value	Source	Rendered on
Invoice / Bill Total	`billings.total_amount`	Finance (Invoices tab), PatientPortal, HistoryStep, BillingDetailDialog, Dashboard, Analytics, FinanceDownloadReports, FYTreatmentRevenue
Amount Paid (per invoice)	`billings.amount_paid`	Finance, PatientPortal, HistoryStep, BillingDetailDialog
Payment Status	`billings.payment_status`	Finance, BillingDetailDialog, PatientPortal, HistoryStep
Single payment row	`billing_payments.*`	Finance (Payment Collection tab), PatientPortal, HistoryStep
Total Billed (period)	Σ `billings.total_amount` filtered by `created_at`	Finance, FinanceReport, Dashboard, Analytics
Collected (period)	Σ `billing_payments.amount` filtered by `payment_date`	Finance (primary), FinanceReport (with `billings.amount_paid` fallback)
Outstanding	Σ `(total_amount − amount_paid)` for non-cancelled	Finance, FinanceReport, Dashboard
Collection %	`Collected / Total Billed`	Finance, FinanceReport
Daily Revenue (7d)	`billing_payments` by `payment_date` (fallback `billings` by `created_at`)	FinanceReport
Payment-method split	`billings.payment_method` × `amount_paid`	FinanceReport
Expense Total	Σ `expenses.amount` by `expense_date`	Finance, Expenses, FinanceDownloadReports
Expense by Category	grouped `expenses`	Finance, Expenses
Purchase Total	Σ `purchases.total_amount` by `purchase_date`	Finance, Purchases
Vendor Outstanding	purchases where `payment_status≠'paid'`	Purchases
Inventory Value	Σ `inventory_items.quantity × cost_price`	Inventory, Finance
Low-stock count	`inventory_items.status='low_stock'`	Inventory, Dashboard
Ortho Paid	Σ `orthodontic_payments.paid_amount`	OrthoTreatment, EMIInsurance
EMI Approved	`emi_insurance_cases.approved_amount`	EMIInsurance
Insurance Claim	`emi_insurance_cases.approved_claim_amount`	EMIInsurance
Deposit Amount	`deposits.deposit_amount`	DepositsSection, Finance (Deposits tab)
Referral Revenue	`referrals.revenue_generated`	Referrals, Dashboard
FY Actual Revenue	`fy_actual_tracking.actual_revenue`	FY pages
FY Actual Profit	`fy_actual_tracking.actual_profit`	FY pages
Variance status	`fy_variance_tracker.variance_status`	FYVariance

8. Calculation Matrix

KPI	Formula	Source	Date field	Filters	Business rules
Total Billed	Σ `billings.total_amount`	`billings`	`created_at`	period, exclude cancelled/void	Counts each invoice once.

KPI	Formula	Source	Date field	Filters	Business rules
Collected	Σ `billing_payments.amount`	`billing_payments`	`payment_date`	period; parent `billings.payment_status` ∉ {cancelled,void}; `amount > 0`	Cash-basis. Includes payments for invoices billed in earlier periods.
Outstanding	Σ (`billings.total_amount` − `billings.amount_paid`)	`billings`	`created_at` (of bill)	non-cancelled, balance > 0	Snapshot from `amount_paid`, not recomputed from payments.
Collection %	`Collected ÷ Total Billed × 100`	derived	mixed (see above)	period	Will be ≠100 even if no debt due to cross-period drift.
Net Profit	`Collected − Expenses` (Finance.tsx) or `Revenue − Expense` (FY actuals)	`billing_payments`, `expenses` / `billings`, `expenses`	mixed	period	**Two definitions exist** (cash vs accrual).
Expenses (period)	Σ `expenses.amount`	`expenses`	`expense_date`	period	Includes auto-mirrored purchase expenses.
Inventory Value	Σ `quantity × cost_price`	`inventory_items`	snapshot	`is_active=true`	Uses last `cost_price` write (no FIFO/LIFO).
Purchase Total	Σ `purchases.total_amount`	`purchases`	`purchase_date`	period	Includes unpaid.
Vendor Paid	Σ `purchases.total_amount` where `payment_status='paid'`	`purchases`	`purchase_date`	period	Mirror also exists in `expenses`.
FY Actual Revenue	Σ `billings.total_amount` for the FY-month	`billings` via `recompute_fy_actuals`	`created_at`	FY month, branch_id (NOT effective — see §11)	Manual override flag honored.
FY Actual Expense	Σ `expenses.amount` for FY-month	`expenses`	`expense_date`	FY month	Categorized into marketing/salary subtotals.
FY Actual Profit	`Revenue − Expense`	derived	FY month	—	Stored, not live.
Variance	`Actual − Target`	`fy_variance_tracker`	—	period	Status = green/yellow/red by 90% threshold.

9. Event Lifecycle

```
sequenceDiagram
  participant P as Patient
  participant W as Workflow UI
  participant DB as Postgres
  participant T as Triggers
  participant F as Finance & FY
  P->>W: Check-in
  W->>DB: patient_checkins
  P->>W: Treatment + Billing
  W->>DB: billings + billing_items
  DB->>T: sync_referral_from_billing
  P->>W: Pay at checkout
  W->>DB: billing_payments
  W->>DB: UPDATE billings.amount_paid, payment_status
  DB->>T: set_billing_payment_dedupe_key
  W->>DB: inventory_items (consume) + inventory_transactions(stock_out)
  Note over DB,F: Finance reads aggregate live
  W->>F: User opens Finance / FY page
  F->>DB: SELECT billings, billing_payments, expenses, purchases
  F->>F: client-side aggregation
  Note over DB,F: FY actuals refreshed on demand
```

```
    W->>DB: rpc(recompute_fy_actuals)
    DB->>DB: write fy_actual_tracking + fy_variance_tracker
```

Parallel lifecycle for vendor purchases:

```
  Purchases.tsx insert → purchases + purchase_items
    → trigger handle_purchase_inventory_update → inventory_items + inventory_transactions(stock_in)
    → trigger sync_purchase_to_expense (when payment_status='paid') → expenses
    → trigger set_purchase_delivery_date (when delivery_status='delivered')
```

10. Single Source of Truth

Concept	Authoritative table	Date axis	Notes
Revenue (accrual)	`billings.total_amount`	`created_at`	Used by FY actuals and Total Billed.
Cash collected	`billing_payments.amount`	`payment_date`	Used by Finance Collected card.
Outstanding	`billings.total_amount − billings.amount_paid`	`created_at`	Snapshot field; not recomputed live from payments.
Expenses	`expenses.amount`	`expense_date`	Auto-mirror from purchases is included.
Inventory value	`inventory_items.quantity × cost_price`	snapshot	—
Inventory ledger	`inventory_transactions`	`created_at`	Append-only.
Vendor payments	`purchases` (`payment_status='paid'`)	`purchase_date`	Also surfaces as a row in `expenses`.
Pharmacy sales	**No dedicated source**	—	Gap. Either reuse `billings+billing_items` with a category flag, or add `pharmacy_sales`.
Deposits	`deposits` + `deposit_invoices`	`deposit_date`	Reconciliation aid, not a money mutation.
Ortho payments	`orthodontic_payments`	`paid_date`	**Not summed into** `billing_payments` — Finance Collected does not include ortho EMIs unless also booked as `billing_payments`.
EMI/Insurance	`emi_insurance_cases.approved_amount/approved_claim_amount`	`approved_at`	Cash event lands in `billing_payments` separately.
Referral revenue	`referrals.revenue_generated`	derived	Auto-updated by `sync_referral_from_billing`.
FY actuals	`fy_actual_tracking`	FY-month	Stale until manual recompute.

11. Architecture Issues

1. **No** `branch_id` **on money tables.** Confirmed by `information_schema`: `billings`, `billing_items`, `billing_payments`, `expenses`, `purchases`, `purchase_items`, `inventory_items`, `inventory_transactions`, `deposits`, `orthodontic_payments`, `emi_insurance_cases`, `lab_orders`, `diagnostic_orders`, `referrals` have no branch column. The `BranchContext` filter on Finance is non-functional for these reads.

2. **Dual "Collected" definitions.** `Finance.tsx` uses `billing_payments` by `payment_date`; `FinanceReport.tsx` falls back to Σ `billings.amount_paid` when `billingPayments` is empty. The two diverge for cross-period payments.

3. **Net Profit ambiguity.** `Finance.tsx` Net Profit uses Collected (cash). FY actual profit uses Revenue (accrual). Same label, different meaning.

4. **No payment status on** `billing_payments`**.** `billing_payments` has no `status` / `is_void` / `refund_of_id` columns. Refunds and corrections cannot be modeled cleanly.

5. **Purchase-to-expense double-counting risk.** `sync_purchase_to_expense` writes an `expenses` row when `payment_status='paid'`. If a user also manually enters that expense (separate flow), it will be counted twice. There is no uniqueness guard beyond `purchase_id`.

6. **Pharmacy sales gap.** `Pharmacy.tsx` does not insert into `billings` for OTC drug sales; pharmacy revenue is invisible to Finance.

7. **FY actuals staleness.** `recompute_fy_actuals` is only called from a "Refresh" button. No `AFTER INSERT/UPDATE` triggers on `billings`/`expenses` keep it fresh.

8. **Outstanding uses snapshot field.** `billings.amount_paid` is the snapshot, not `Σ billing_payments`. If a payment is hand-edited or deleted, snapshot can drift from the ledger.

9. **Ortho payments are orphan to Finance.** `orthodontic_payments` is not unioned into Collected. Finance under-reports cash if ortho EMIs are not also booked through `billing_payments`.

10. **Missing indexes** likely on `billing_payments(payment_date)`, `expenses(expense_date)`, `purchases(purchase_date)` — to be confirmed; client-side aggregation will scan-table at growth.

11. **No audit log.** No `audit_log` table for money mutations. Refund forensics rely on `created_at` only.

12. **Dead code paths.** `FinanceReport.tsx` collected-fallback branch is dead in practice (Finance always loads payments) but kept for safety; document as deprecated.

12. QA Testing Matrix

Action	DB writes	Expected UI updates	Expected reports/dashboards
Create new bill in Workflow	`billings` + `billing_items` insert	PaymentStep shows invoice, PatientPortal & HistoryStep show new bill	Total Billed ↑, Outstanding ↑
Add full payment	`billing_payments` insert + `billings.amount_paid+payment_status` update	Bill marked Paid; PatientPortal shows transaction	Collected ↑, Outstanding ↓, Collection % ↑
Add partial payment	same as above; `payment_status='partial'`	Partial chip on bill	Collected ↑ by partial; Outstanding ↓ by partial
Edit a billing item amount	`billing_items.update` + `billings.subtotal/total_amount` recomputed	Bill total redrawn	Total Billed and Outstanding adjust
Delete a bill	`billing_items.delete` + `billings.delete`	Bill disappears	Total Billed ↓, Outstanding ↓; Collected unchanged unless payments cascade
Create purchase, mark paid	`purchases+purchase_items` insert → triggers add `inventory_items+inventory_transactions(stock_in)+expenses`	Purchases list updates; Inventory qty ↑; Expenses list shows mirrored row	Finance Expenses ↑; Inventory value ↑
Create purchase, unpaid	`purchases+purchase_items` only; no `expenses` row	Inventory still ↑ via trigger	Expenses unchanged
Mark unpaid purchase as paid	trigger inserts mirrored `expenses` row	Expenses list updates	Finance Expenses ↑
Add manual expense	`expenses.insert`	Expenses list updates	Finance Expenses ↑
Record a deposit	`deposits.insert` + `deposit_invoices.insert` per linked bill	DepositsSection list updates	Reconciliation report updates; no change to Collected
Record an ortho payment	`orthodontic_payments.insert`	Ortho page updates	**Finance Collected unchanged** (gap)
Approve EMI/Insurance	`emi_insurance_cases.update`	EMI page updates	No automatic Finance impact
Run FY actuals refresh	`fy_actual_tracking` + `fy_variance_tracker` upsert	FY dashboards redraw	Variance status recolors

Action	DB writes	Expected UI updates	Expected reports/dashboards
Switch branch	Branch context updates	Most clinical pages refilter	**Finance shows same numbers** (no branch_id on money tables)

13. Recommendations

1. **Add** `branch_id` **to every money table** with backfill, RLS update, and required filter on every Finance query. Without this, multi-branch finance is unsupported.

2. **Add lifecycle columns to** `billing_payments` — `status` (`active`/`void`/`refund`), `voided_at`, `voided_by`, `refund_of_id` — and treat them in all aggregations.

3. **Promote** `billing_payments` **to the single Outstanding source.** Compute `amount_paid` from `Σ billing_payments.amount` via a trigger so the snapshot never drifts.

4. **Unify Collected / Net Profit definitions.** Pick cash-basis OR accrual-basis explicitly and label all cards accordingly.

5. **Trigger FY recompute on** `billings`**/**`expenses` **mutations** (debounced) instead of manual refresh; or schedule a nightly job.

6. **Close the Pharmacy gap** — route OTC sales through `billings` with a `source='pharmacy'` discriminator.

7. **Union ortho/EMI cash events into** `billing_payments` so Finance Collected is exhaustive; keep ortho schedule table for installment plan management only.

8. **Add a vendor-ledger view** (`v_vendor_ledger`) joining `purchases` + `expenses` to remove double-count risk and give finance a single AP view.

9. **Add indexes**: `billing_payments(payment_date)`, `expenses(expense_date)`, `purchases(purchase_date)`, `billings(created_at)`, plus `(branch_id, <date>)` composites after #1.

10. **Add** `finance_audit_log` (table + trigger on every money table) capturing user, before/after, source page. Critical for HMS/ERP compliance.

11. **Move aggregation server-side** via Postgres views or RPCs (`v_finance_kpis(period_start, period_end, branch_id)`) so UI does not scan thousands of rows.

12. **Document Collected vs Billed drift** in the UI (already done as a tooltip on Finance.tsx).

14. Appendices

14.1 Glossary

- **Accrual basis** — revenue when billed.

- **Cash basis** — revenue when received.

- **Cross-period drift** — payments in period N for bills issued in period N–1.

- **Authoritative source** — the table that the rest of the system should derive from.

14.2 Regeneration runbook

1. Re-read these files: `Finance.tsx`, `FinanceReport.tsx`, `PaymentStep.tsx`, `Expenses.tsx`, `Purchases.tsx`, `Inventory.tsx`, `Pharmacy.tsx`, `DepositsSection.tsx`, `EMIInsurance.tsx`, `OrthoTreatment.tsx`, `useFYData.ts`.

2. Re-pull schema for money tables: `SELECT table_name, column_name FROM information_schema.columns WHERE table_schema='public' AND table_name IN (...)`.

3. Re-run sanity SQL: `SELECT date_trunc('month', payment_date) m, SUM(amount) FROM billing_payments GROUP BY 1 ORDER BY 1`.

4. Update Section 11 with any new triggers / tables.

5. Bump the version footer below and regenerate the PDF/DOCX from the same Markdown.

14.3 Version footer

Field	Value
Version	v1.0
Generated	2026-06-30
Author	Lovable Finance Audit
Status	Read-only audit, no implementation changes

15. Data Lineage Matrix

Field-level trace from DB column to user-facing surface.

Metric	DB Source	Business Logic	Hook / Loader	Component	Dashboard	Reports	Export
Invoice Amount	`billings.total_amount`	`SUM(billing_items.total_price)` at insert	`Finance.tsx` direct `supabase.from('billings').select()`	`BillingDetailDialog`, `FinanceReport`	Finance, FY Finance	FinanceReport, SalesReport	`exportAnalytics.ts`
Amount Paid	`billings.amount_paid`	Sum of `billing_payments.amount` per bill, updated in `PaymentStep`	`Finance.tsx`, `PatientPortal`	`PaymentStep`, `BillingDetailDialog`	Finance, Dashboard	FinanceReport	DownloadReports
Outstanding	`billings.total_amount - billings.amount_paid` (status ≠ cancelled)	derived in component	`Finance.tsx`	KPI card	Finance, FY Finance	FinanceReport	FinanceDownloadReports
Collected (cash)	`billing_payments.amount` by `payment_date`	SUM filtered to period	`Finance.tsx`	KPI card "Collected"	Finance	FinanceReport (`billingPayments`-path), GrowthReport	FinanceDownloadReports
Revenue (accrual)	`billings.total_amount` by `created_at`	SUM per period	`useFYData.ts` for FY; direct query elsewhere	KPI card "Total Revenue"	Finance, FY Finance, SuperDashboard	SalesReport, FinanceReport	exportAnalytics
Expense Amount	`expenses.amount` by `expense_date`	SUM per category	`Expenses.tsx`, `Finance.tsx`	Expenses table, KPI card	Finance, FY Profit	FinanceReport	DownloadReports
Purchase Total	`purchases.total_amount` by `purchase_date`	SUM	`Purchases.tsx`	Purchases table	Purchases page	—	—
Inventory Value	`inventory_items.quantity * cost_price`	SUM over rows	`Inventory.tsx`, `InventoryStatsCards`	KPI card	Inventory page	—	—
Vendor Payment	`purchases` (status=paid) → auto `expenses` row	Trigger `sync_purchase_to_expense`	`Purchases.tsx`	Vendor table	Finance (as expense)	FinanceReport	—
Cash Collection	`billing_payments.amount`	same as Collected	`Finance.tsx`	KPI	Finance	FinanceReport	FinanceDownloadReports
Net Profit	Revenue – Expenses	derived	`Finance.tsx`, `useFYData.ts`	KPI	Finance, FY Profit	GrowthReport	—
FY Actual Revenue	`fy_actual_tracking.actual_revenue`	populated by `recompute_fy_actuals` RPC	`useFYTable.ts`	FY tables	FY Finance, FY Variance	—	—

Metric	DB Source	Business Logic	Hook / Loader	Component	Dashboard	Reports	Export
Ortho Paid	`orthodontic_payments.amount`	SUM per case	direct query in `OrthoTreatment.tsx`	Ortho case detail	—	—	—
Deposit Amount	`deposits.amount`	SUM open deposits	`DepositsSection`	Finance	—	—	—

16. SQL / Data Access Audit

All access is via `@/integrations/supabase/client` → PostgREST. No raw SQL outside DB functions.

Surface	Tables	Joins (foreign-table syntax)	Filters	Date Field	Branch Filter?	Notes
`Finance.tsx` collected	`billing_payments`	none	`payment_date BETWEEN :from AND :to`	`payment_date`	■ (no column)	Branch dropdown is cosmetic
`Finance.tsx` revenue	`billings`	`patients(name)`, `billing_items(*)`	`created_at BETWEEN :from AND :to`, `payment_status != 'cancelled'`	`created_at`	■	
`Finance.tsx` outstanding	`billings`	none	`payment_status IN ('pending','partial')`	`created_at`	■	
`Expenses.tsx`	`expenses`	`expense_categories(name)`, `expense_subcategories(name)`	`expense_date BETWEEN ...`	`expense_date`	■	
`Purchases.tsx`	`purchases`	`suppliers(name)`, `purchase_items(*, inventory_items(name))`	`purchase_date BETWEEN ...`	`purchase_date`	■	Triggers mirror to expenses+inventory
`Inventory.tsx` value	`inventory_items`	`inventory_categories(name)`, `branches(name)`	none	n/a	■ via `branch_id`	Only money-adjacent table with branch_id
`PaymentStep.tsx` insert	`billings`, `billing_items`, `billing_payments`, `inventory_items`, `inventory_transactions`	n/a	n/a	n/a	n/a	Multi-step write; not atomic
`useFYData.ts`	`fy_master_plan`, `fy_monthly_targets`, `fy_actual_tracking`, `fy_variance_tracker`	none	`financial_year`, `branch_id`	`month`	■	FY tables DO have branch_id
`OrthoTreatment.tsx`	`orthodontic_cases`, `orthodontic_payments`, `orthodontic_visits`	`patients(name)`	`case_id`	`payment_date`	■	
`DepositsSection.tsx`	`deposits`, `deposit_invoices`	`patients(name)`	`status`	`created_at`	■	

Indexes worth confirming: `billing_payments(payment_date)`, `expenses(expense_date)`, `purchases(purchase_date)`, `billings(created_at, payment_status)`. None of these were verified in this audit — Recommendation §13.

Performance notes: Finance.tsx fetches the full month of `billing_payments` + `billings` + `billing_items` + `expenses` on every render. No pagination. Acceptable today; will degrade past ~50k rows/month.

17. Impact Matrix

Per-action reach across the system.

17.1 Patient pays ■500 at checkout

```
PaymentStep.handlePayment(500)
  ■■■ INSERT billing_payments (amount=500, payment_date=workflowDate)
       ■■■ TRIGGER set_billing_payment_dedupe_key   (dedup key generated)
       ■■■ UPDATE billings.amount_paid += 500, payment_status (paid|partial)
            ■■■ TRIGGER sync_referral_from_billing  (if patient came via referral)
Visible in:
  • Finance.tsx       — Collected ↑, Outstanding ↓
  • FinanceReport     — Daily Revenue chart ↑ on payment_date
  • Dashboard         — Today's Collection ↑
  • PatientPortal     — Payment history row appears
  • BillingDetailDialog — bill marked paid/partial
  • GrowthReport      — month KPI ↑
  • FY Actuals        — only after manual recompute_fy_actuals
```

17.2 Purchase marked PAID

```
Purchases.tsx update purchases.status='paid'
  ■■■ TRIGGER sync_purchase_to_expense   → INSERT expenses (auto)
  ■■■ TRIGGER handle_purchase_inventory_update → UPDATE inventory_items.quantity, INSERT inventory_transactions
Visible in:
  • Purchases page     — row turns green
  • Expenses page      — auto-expense row visible (■ double-count risk if user also added manual)
  • Inventory          — stock level ↑
  • Finance KPI Expense — ↑
  • FY Profit          — only after recompute
```

17.3 Manual Expense added

```
INSERT expenses (amount, expense_date, category_id)
  ■■■ TRIGGER generate_expense_number  → auto EXP-YYYY-NNNN
Visible in:
  • Expenses page, Finance KPI, FinanceReport, FY Profit (after recompute)
```

17.4 Refund (current pattern: negative payment row)

```
INSERT billing_payments (amount=-X, payment_date=today)
  ■■■ UPDATE billings.amount_paid -= X
■ No void/refund status. Reconciliation relies on negative-amount detection.
```

17.5 Deposit recorded

```
INSERT deposits (amount, status='open')
  ■■■ later: INSERT deposit_invoices linking to billings (allocation)
Visible only on Finance > Deposits section. Not in Finance KPIs.
```

17.6 Pharmacy sale

```
Currently: inventory_items.quantity -= n via Pharmacy.tsx
NO automatic billings/billing_payments row created.
■ Pharmacy revenue invisible to Finance unless user manually creates a bill.
```

17.7 Ortho payment

```
INSERT orthodontic_payments (case_id, amount, payment_date)
Visible only on OrthoTreatment case detail.
■ Not aggregated into Finance.Collected.
```

17.8 EMI / Insurance disbursement

```
UPDATE emi_insurance_cases.disbursed_amount, .status
Visible only on EMI/Insurance page. Not aggregated into Finance KPIs.
```

18. Version Metadata

Field	Value
Document Version	**v2.0**
Supersedes	v1.0 (`docs/Finance_Architecture_Audit.md`)
Generation Date	2026-06-30
Project Version (package.json)	0.0.0
Git Commit	ad3055ff
Database	Postgres 15 (Lovable Cloud / Supabase)
Environment	Production schema, Preview app
Generated By	Lovable Plan-Mode Audit
Tooling	code--view, rg, supabase--read_query (SELECT only)

19. Regeneration Guide

After each release that touches finance code or schema:

1. **Re-scan code** — `rg "billing_payments|billings|expenses|purchases|inventory_items|orthodontic_payments|deposits" src`.

2. **Re-pull schema** — `SELECT table_name, column_name FROM information_schema.columns WHERE table_schema='public' AND table_name IN ('billings','billing_payments','expenses','purchases','purchase_items','inventory_items','inventory_transactions','deposits','orthodontic_payments','emi_insurance_cases')`.

3. **Sanity checks** —

```sql
SELECT date_trunc('month', payment_date) m, SUM(amount) FROM billing_payments GROUP BY 1 ORDER BY 1;

SELECT date_trunc('month', expense_date) m, SUM(amount) FROM expenses GROUP BY 1 ORDER BY 1;
```

4. **Update Sections 11, 15, 16, 17** with any new tables/triggers/columns.

5. **Bump Version Metadata** (§18): version, date, git SHA.

6. **Re-render artifacts** — run the scripts under `docs/diagrams/` and re-emit `Finance_Architecture.pdf` and `.docx`.

7. **Append to** `11_Release_Notes.md`.

8. **Update** `docs/VERSIONS.json` with the new entry.